Day 035 | Unit 03 | Teacher plan

### Integers and Rational Numbers

# Order Negative Fractions with Benchmarks

Standards: 6.NS.C.6c, 6.NS.C.7a, 6.NS.C.7b

Learning target: I can order Negative Fractions with Benchmarks, and explain and check my reasoning.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

## 0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

## 5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Mark zero and equally spaced whole-number ticks. Partition each interval into equal parts for the denominator. Negative values are left of zero on a horizontal line and below zero on a vertical line. A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions.

## 17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

## 42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

## Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

## Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 1, 24

Day 035 | Unit 03 | Lesson

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## 01 / Learn

Mark zero and equally spaced whole-number ticks. Partition each interval into equal parts for the denominator. Negative values are left of zero on a horizontal line and below zero on a vertical line. A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions.

## Worked example 1

Draw a horizontal or vertical number line. Plot and label -8, 0, and 6.

1. Mark zero and equally spaced whole-number ticks.

2. Partition each interval into equal parts for the denominator.

3. Negative -8 is left of or below 0

4. 6 is right of or above 0.

5. Use equal intervals.

6. State the result in the form requested: -8 < 0 < 6; unit steps.

Answer: -8 < 0 < 6; unit steps

## Worked example 2

Temperatures are -5 °C and -3 °C. Select all true comparisons.
A. -5 is warmer than -3.
B. -5 < -3
C. -3 lies to the right of -5 on a number line.
D. -5 > 0

1. A number farther right is greater.

2. Among negatives, the number closer to zero is greater.

3. Thus -3 is warmer and is farther right.

Answer: B, C - -5 < -3; -3 lies to the right of -5 on a number line.

# Day 035 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

### 01 / Constructed response

Draw a horizontal or vertical number line. Plot and label -8, 0, and 4.

### 02 / Constructed response

Compare -3 ___ -3 using <, >, or =. Sketch a number line to justify.

### 03 / Constructed response

Draw a horizontal or vertical number line. Plot and label -6, 0, and 2.

### 04 / Constructed response

Compare -2 ___ -1/2 using <, >, or =. Sketch a number line to justify.

### 05 / Constructed response

Draw a horizontal or vertical number line. Plot and label -2/3, 0, and 4/3.

### 06 / Constructed response

Compare -3/5 ___ -4/3 using <, >, or =. Sketch a number line to justify.

## 07 / Constructed response

Draw a horizontal or vertical number line. Plot and label -5/4, 0, and 3/4. Name the step size of your scale and explain the placement.

## 08 / Multiple select

Temperatures are -6 °C and -4 °C. Select all true comparisons.
A. -6 is warmer than -4.
B. -6 < -4
C. -4 lies to the right of -6 on a number line.
D. -6 > 0

## 09 / Graphing and two-part response

On a number line, consecutive integer ticks are 1 unit apart. Plot -13/2, -6, and 0. Part A: name the leftmost point. Part B: give the midpoint between -7 and -6.



Choose and label an equal-interval scale.

## 10 / Constructed response

Temperatures are -7/5°C and -2/5°C. Write an inequality comparing them and explain which is warmer.

Day 035 | Unit 03 | Answer key

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## 1. -8 < 0 < 4; unit steps

Negative -8 is left of or below 0; 4 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

## 2. =

-3 = -3; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

## 3. -6 < 0 < 2; unit steps

Negative -6 is left of or below 0; 2 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

## 4. <

-2 < -1/2; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

## 5. -2/3 < 0 < 4/3; use steps of 1/3

Negative -2/3 is left of or below 0; 4/3 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

## 6. >

-3/5 > -4/3; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

## 7. -5/4 < 0 < 3/4; use steps of 1/4

Negative -5/4 is left of or below 0; 3/4 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

## 8. B, C - -6 < -4; -4 lies to the right of -6 on a number line.

Among negatives, the number closer to zero is greater. Thus -4 is warmer and is farther right.

6.NS.C.7a, 6.NS.C.7b | Multiple select

## 9. Leftmost -13/2; midpoint -13/2

-13/2 = -6 - 1/2 lies halfway between -7 and -6, left of both -6 and 0.

6.NS.C.6c | Graphing and two-part response

## 10. -7/5 < -2/5; -2/5°C is warmer

The greater value is farther right on a number line, even when both are negative.

6.NS.C.7a, 6.NS.C.7b | Constructed response